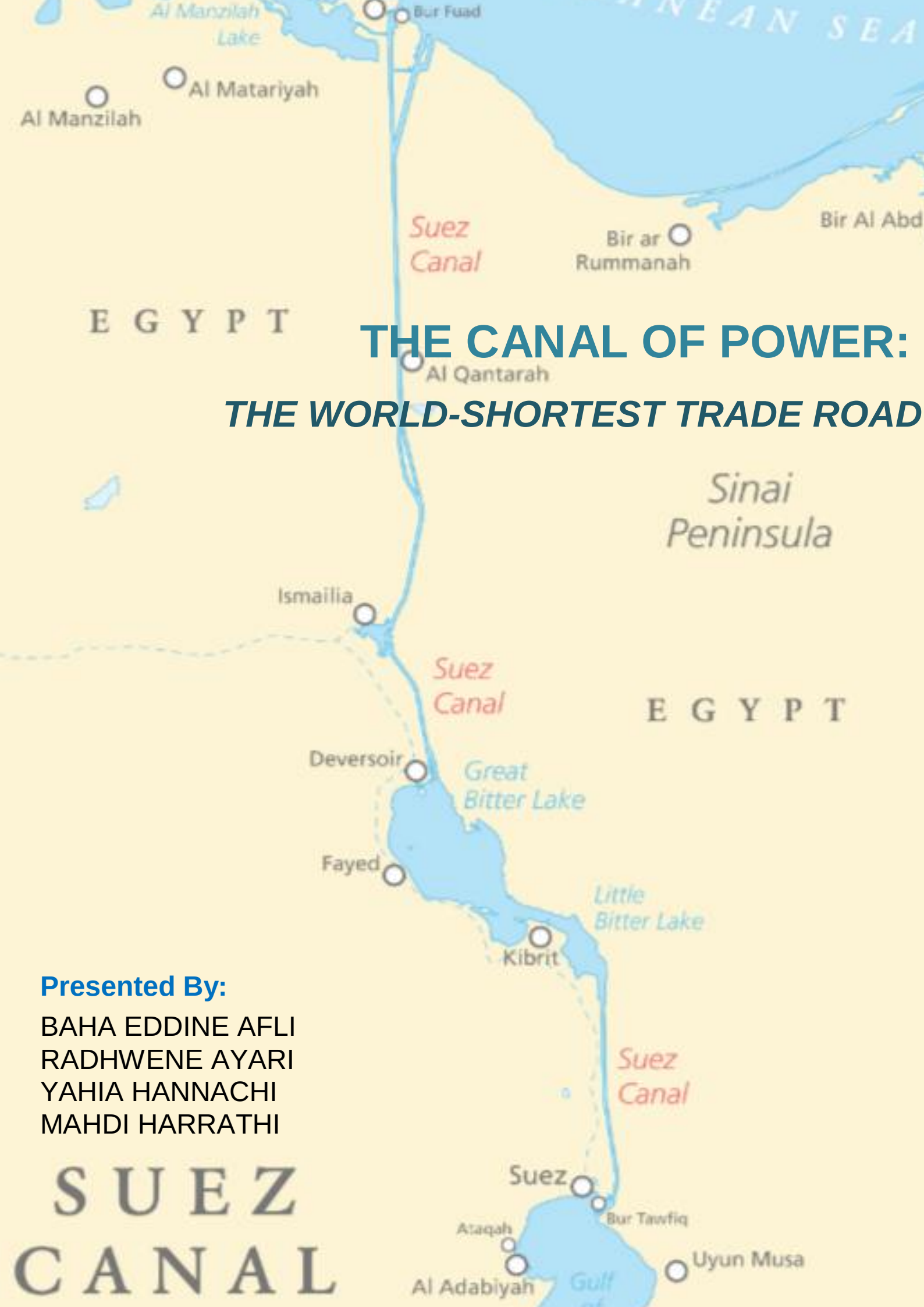


THE CANAL OF POWER:

THE WORLD-SHORTEST TRADE ROAD

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1. Industry & Context

This project focuses on the global logistics and maritime transport industry, with specific emphasis on the Suez Canal, one of the most critical trade routes worldwide. Approximately 12% of global trade passes through the canal, connecting Asia, Europe, and North Africa, including Tunisia. Any disruption such as congestion, weather conditions, accidents, or geopolitical risks can result in shipment delays, increased transportation costs, and inventory shortages.

2. Business Problem

Logistics companies and importers face unpredictable shipment delays, increased freight costs, limited visibility into congestion risks, and reactive decision-making processes. The absence of real-time analytical insights prevents proactive planning and effective risk mitigation strategies.
Problem Statement:
How can logistics operators predict delays and cost overruns for shipments passing through the Suez Canal and optimize routing and inventory decisions?

3. BI Solution

The proposed Business Intelligence solution consists of interactive dashboards that monitor vessel traffic, congestion patterns, and shipment delays. The system measures the financial impact of disruptions, identifies high-risk periods and routes, and supports data-driven rerouting and inventory buffer decisions.

4. Dataset Availability

The dataset will include between 20,000 and 100,000 shipment records. Data sources include Kaggle global shipping datasets, AIS vessel tracking samples, global trade and shipping delay datasets, and public Suez Canal traffic statistics. Data will be localized to represent shipments originating from Asia and Europe and arriving at destinations including Tunisian ports such as Rades and Sfax.

5. Data Model

Fact Table: Shipments_Fact
- Shipment_ID
- Transit_Time_Hours

- Delay_Hours
- Freight_Cost
- Fuel_Cost
- Inventory_Delay_Cost
- Shipment_Count

Dimension Tables:
Date_Dim, Vessel_Dim, Route_Dim, Risk_Dim

6. Key Performance Indicators

Average Transit Time, Average Delay Hours, On-Time Delivery Rate, Cost per Shipment, Delay Cost Impact (%), Congestion Index, Risk Exposure Score, Percentage of Shipments Rerouted, Year-over-Year Transit Time Change.

7. Analytical Questions

Which characteristics make the Suez Canal a critical chokepoint in international supply chains?
When are shipment delays through the Suez Canal most frequent (by month, week, or day)?
Which vessel types (container, oil tanker, bulk carrier) experience the longest delays?
How do shipment delays affect overall freight costs for different routes?
What is the cost difference between using the Suez Canal versus alternative routes (e.g., Cape of Good Hope)?
Which months or seasons exhibit the highest congestion levels?
How do delays specifically impact imports arriving at Tunisian ports such as Rades and Sfax?
Can delay patterns be predicted using historical congestion and traffic data?
Which risk factors (weather, accidents, congestion, geopolitical events) contribute most to shipment delays?
How much inventory buffer is required at ports to mitigate the impact of high-risk periods?
What are the year-over-year trends in congestion and average transit times for shipments through the Suez Canal?

8. Dashboards

The BI solution includes an Executive Summary Dashboard, a Deep Dive Analysis Dashboard, and advanced visualizations such as geographical route maps, delay forecasts, and shipment-level drill-through views.

9. Conclusion

This project presents a realistic and high-impact BI use case addressing global supply chain challenges through the Suez Canal.